

Week 11

23-4-2024

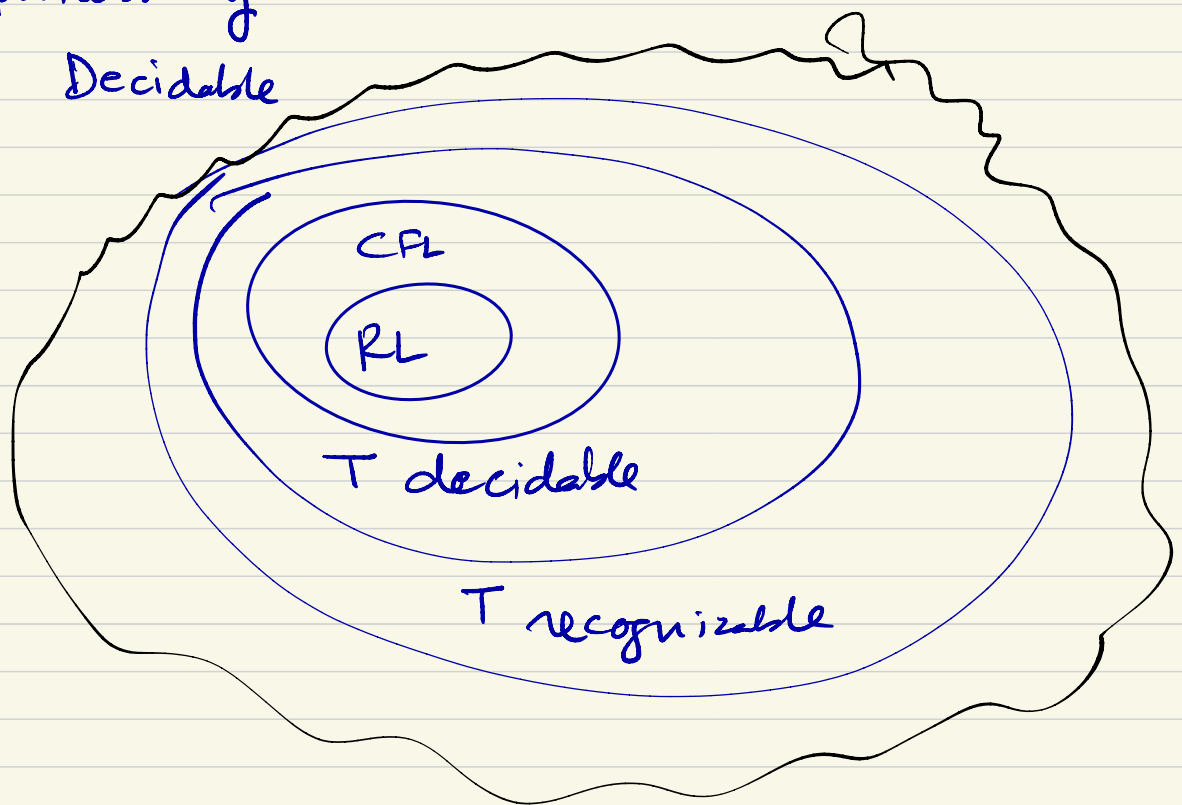


23-4-2024

Time Complexity

Computability

Decidable



Efficiency

$A = \{0^k 1^k \mid k \geq 0\}$. is A decidable?

$x \in \Sigma^*$ $\Sigma = \{0, 1\}$

$x \in A$ or $x \notin A$

which TM decides A ?

x of size n $|x| = n$
 $\downarrow$
 $M_1, M_2, M_3, \dots$

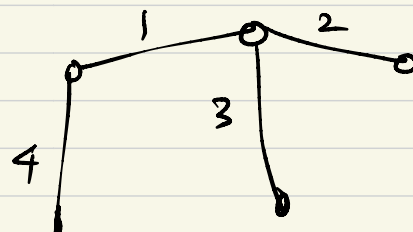
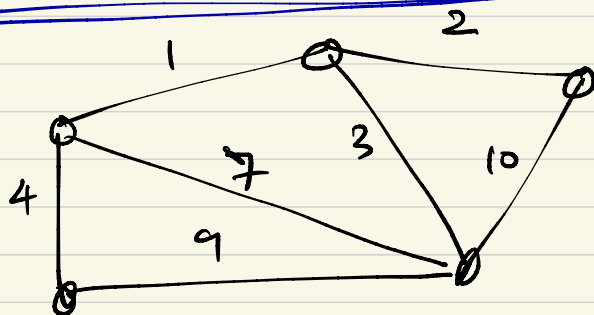
measuring the time complexity

① Optimization Problems

Sorting, Searching, ...

② Decision Problems

Sorting



$X = \langle 1, 5, 6, 9, 10, 12 \rangle$

my Question: is X sorted?

yes

$Y = \langle 3, 2, 9, 0 \rangle$

no

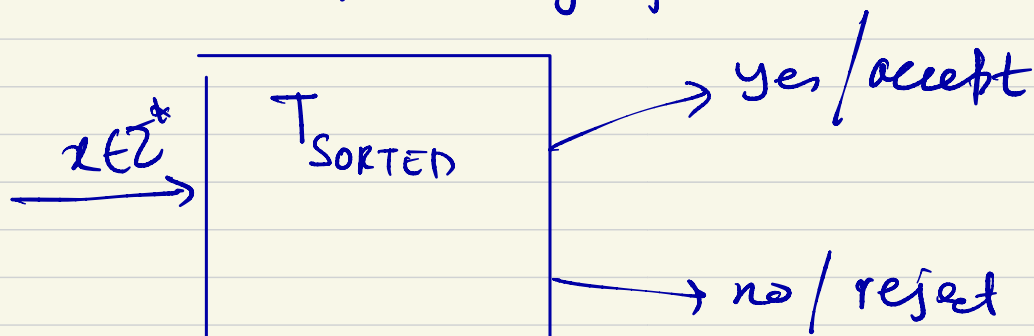
is Y sorted?

$\text{SORTED} = \{ \langle X \rangle \mid X \text{ is a sorted sequence} \}$

$\Sigma = \{0, 1, \dots, 9\}$
 $x \in \Sigma^*$

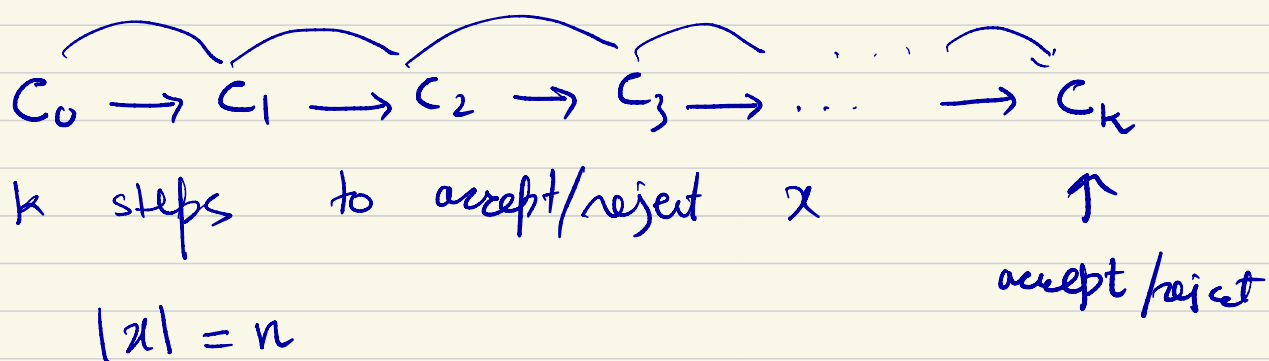
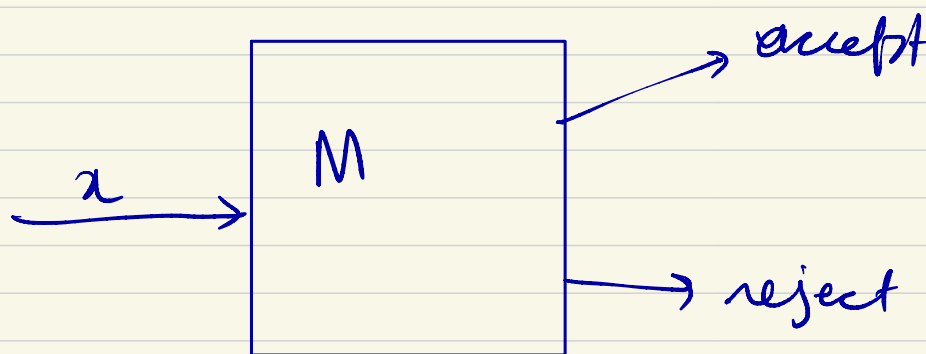
does $x \in \text{SORTED}$?

membership Query problem.

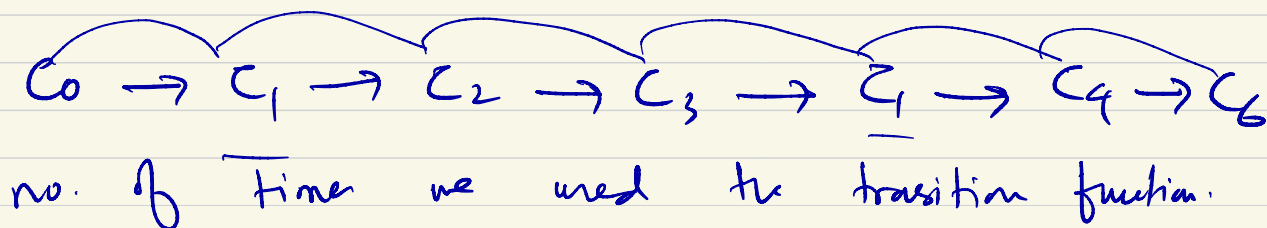


Definition

Let M be a deterministic TM that halts on all input. The "running time" or "time complexity" of M is the function $f: \mathbb{N} \rightarrow \mathbb{N}$, where $f(n)$ is the maximum no. of steps that M uses on any input of length n . If $f(n)$ is the running time of M , we say M runs in time $f(n)$ and M is an $f(n)$ -time Turing Machine.



$f(n)$ is the maximum no. of steps by M on any input of length n .



$f(n)$ is the max no. of steps
required by M to
solve the input of size n .

$$|x| = 100$$

p if p is prime

$$\underline{2}, \underline{3}, \underline{5}, \underline{7}, \underline{11}, \underline{13}, \dots, p-1$$

p with 100 digits

$$p = d_1 d_2 \dots \underline{\underline{d_{100}}}$$

$$d_{100} = 0, 2, 4, 6, 8,$$

M

$$f_M(n) = 2n^2 - n + 3$$

M_2

$$f_{M_2}(n) = 2n^2 - 2n + 5$$

$\vdots$

Notations. to compare different functions.

Oh notations.

Definition.

$\mathbb{R}^+$ means non-negative real no.s
 $\mathbb{R}^{\geq 0}$

Let f and g be functions, $f, g: \mathbb{N} \rightarrow \mathbb{R}^+$.
we say $f(n) = O(g(n))$ if
positive integers c and n_0 exist s.t.
for every $n \geq n_0$.

$$f(n) \leq c \cdot g(n).$$

where $f(n) = O(g(n))$ $f(n) \in O(g(n))$
"f(n) is in $O(g(n))$ "

we say that $g(n)$ is an upper bound
for $f(n)$, or more precisely, that
 $g(n)$ is an asymptotic upper bound for $f(n)$
to emphasize that we are ignoring
constant factors.